

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 13-2024: A 27-Year-Old Man with Leg Weakness

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and Peter M. Sadow, M.D., Ph.D.

PRESENTATION OF CASE

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CME



Dr. Jamie Aron (Emergency Medicine): A 27-year-old man was evaluated at this hospital because of leg weakness.

The patient had been in his usual state of health until 3 days before the current presentation, when myalgias developed in the upper legs. He initially attributed the pain to increased physical activity while at work in construction. When he awoke on the morning of the current presentation, he could not lift his legs to move in bed or to stand up. He also had weakness in the arms, although it was less pronounced than the weakness in the legs. He asked his family to bring him to the emergency department of this hospital for evaluation.

In the emergency department, the patient reported ongoing weakness. He had no history of trauma, back pain, or saddle anesthesia. There were no changes in sensation. He had no other relevant medical history, and he took no prescribed medications. He had moved to Boston from El Salvador 5 months before the current presentation.

After arriving in Boston, the patient had begun to have increasing difficulty with falling asleep and staying asleep, which he had attributed to stress. He had had loose stools but a normal appetite and was eating normally with no abdominal pain, nausea, or vomiting; he estimated a weight loss of approximately 5 kg since moving to Boston. In addition, he reported that he had had trembling in his hands for several months before the current presentation, which he had also attributed to stress. Four days before the current presentation, he began taking an herbal supplement (sold under the brand name Nervitran) that was intended to reduce stress. He had no known allergies. He drank alcohol occasionally, had never smoked tobacco, and did not use illicit drugs. There was no family history of hypertension or liver disease; a maternal aunt had had an episode of weakness, but the patient was not aware of a diagnosis.

On examination, the temporal temperature was 36.2°C, the blood pressure 144/79 mm Hg, the pulse 96 beats per minute, the respiratory rate 18 breaths per

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	On Presentation
Sodium (mmol/liter)	135–145	141
Potassium (mmol/liter)	3.4–5.0	1.8
Chloride (mmol/liter)	98–108	106
Carbon dioxide (mmol/liter)	23–32	23
Urea nitrogen (mg/dl)	8–25	17
Creatinine (mg/dl)	0.60–1.50	0.53
Glucose (mg/dl)	70–110	141
Anion gap (mmol/liter)	3–17	12
Calcium (mg/dl)	8.5–10.5	10.4
Magnesium (mg/dl)	1.7–2.4	1.8
Alkaline phosphatase (U/liter)	30–100	156
Alanine aminotransferase (U/liter)	7–33	67
Aspartate aminotransferase (U/liter)	9–32	54
Bilirubin (mg/dl)		
Total	0.0–1.0	1.3
Direct	0.0–0.4	0.3
International normalized ratio	0.9–1.1	1.1
Albumin (g/dl)	3.3–5.0	4.1
Creatine kinase (U/liter)	60–400	382
Aldolase (U/liter)	<7.7	9.9
Venous blood gases		
pH	7.30–7.40	7.29
Partial pressure of carbon dioxide (mm Hg)	38–50	56
Partial pressure of oxygen (mm Hg)	35–50	32

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for magnesium to millimoles per liter, multiply by 0.4114. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

minute, and the oxygen saturation 99% while the patient was breathing ambient air. The body-mass index (BMI; the weight in kilograms divided by the square of the height in meters) was 21.6. Conjunctival injection was present in both eyes. Muscle strength was 4/5 on abduction of both shoulders and on flexion and extension of the elbows, wrists, and fingers. Muscle strength was 5/5 on knee flexion and extension and 3/5 on hip flexion, ankle dorsiflexion, and plantar flexion. Sensation to light touch and temperature was normal. Patellar reflexes were 3+. There was a low-amplitude, high-frequency fine tremor

in the hands, which was more prominent in the left hand than in the right hand. The remainder of the examination was normal.

The complete blood count was normal, as were the results of tests of coagulation. A blood toxicology screen was negative for salicylates, acetaminophen, ethanol, and tricyclic antidepressant medications. The blood level of aspartate aminotransferase was 56 U per liter (reference range, 10 to 40), the alanine aminotransferase level 65 U per liter (reference range, 10 to 55), the alkaline phosphatase level 137 U per liter (reference range, 45 to 115), the total bilirubin level

1.3 mg per deciliter (22 μ mol per liter; reference range, 0.0 to 1.0 mg per deciliter [0 to 17 μ mol per liter]), the potassium level 1.8 mmol per liter (reference range, 3.4 to 5.0), the creatine kinase level 382 U per liter (reference range, 60 to 400), and the aldolase level 9.9 U per liter (reference value, <7.7); other laboratory test results are shown in Table 1. A urinalysis was not performed. An electrocardiogram showed sinus tachycardia, incomplete right bundle-branch block, and nonspecific ST-segment and T-wave abnormalities. Potassium chloride was administered orally at a dose of 60 meq. Three hours later, the blood level of potassium was 4.2 mmol per liter, and the weakness in the patient's arms and legs had abated.

Additional diagnostic tests were performed.

DIFFERENTIAL DIAGNOSIS

Dr. Andrew S. Allegretti: This 27-year-old previously healthy man presented with a 3-day history of myalgias and proximal muscle weakness that had lasted for several hours, and he was found to have severe hypokalemia. The differential diagnosis for proximal muscle weakness is broad. The presence of severe hypokalemia in combination with an abatement of symptoms after potassium repletion suggests that hypokalemia is the most likely cause of the muscle weakness in this patient. Therefore, I will focus my differential diagnosis on the possible causes of hypokalemia, while being mindful that the most likely diagnosis will also need to account for the patient's more subacute symptoms of weight loss, diarrhea, tremor, and insomnia.

POTASSIUM HOMEOSTASIS

Potassium is the most abundant cation in the body, with a total body content of nearly 3500 mmol. Ninety-eight percent of the total body content of potassium is found in the intracellular fluid owing to the sodium–potassium pump (Na^+/K^+ -ATPase), which actively transports it across cell membranes and leaves only about 60 mmol of potassium available in the extracellular space. A typical Western diet includes approximately 100 mmol of potassium per day. To handle this load without dangerous postprandial increases in the blood potassium level, the body must regulate the level of extracellular potassium quickly and

efficiently. This occurs by means of an intracellular shift through the Na^+/K^+ -ATPase, which is stimulated by catecholamines and insulin. In the long term, approximately 90% of ingested potassium is excreted in the urine through the distal nephron, with the remainder excreted in the stool and through the skin. This patient reported eating normally, although we were not told whether he consumed a typical Western diet. Regardless, it is unusual for hypokalemia to develop in patients as a result of inadequate dietary intake alone because the urinary excretion of potassium can be reduced to as low as 20 mmol per day. Therefore, the next step in this case should focus on determining whether the cause of this patient's hypokalemia is related to renal losses, nonrenal losses, or intracellular shifts.¹

The quantification of urinary potassium excretion is generally the first step in diagnostic testing for hypokalemia (Fig. 1). The use of either a 24-hour urine collection or a spot urine sample is preferred over the use of the transtubular potassium gradient.²⁻⁴ A urinary potassium excretion of at least 20 mmol per day (or ≥ 15 mmol per liter in a spot urine sample) is suggestive of renal potassium wasting. Conversely, an excretion of less than 20 mmol per day (or <15 mmol per liter in a spot urine sample) suggests that the hypokalemia is due to nonrenal losses or transcellular shifts. It is important to note the limitations of tests to quantify urinary potassium. First, the use of diuretic agents confounds the assessment of urinary potassium. Second, renal potassium excretion is dependent on the delivery of sodium to the distal nephron so that sodium can be reabsorbed in exchange for potassium. In patients with impaired delivery of sodium to the distal nephron (e.g., severe intravascular volume depletion), a low urinary sodium level makes the interpretation of the urinary potassium level less useful. Finally, a high volume of urine can result in a low potassium level in a spot urine sample despite renal potassium wasting.^{5,6} Normalizing the potassium level in a spot urine sample with urine creatinine can help offset the effects of high urine volume. Information that would have quantified urinary potassium excretion was not provided in this case, so I will next consider renal potassium loss, nonrenal potassium loss, and intracellular shifts.

RENAL POTASSIUM LOSS

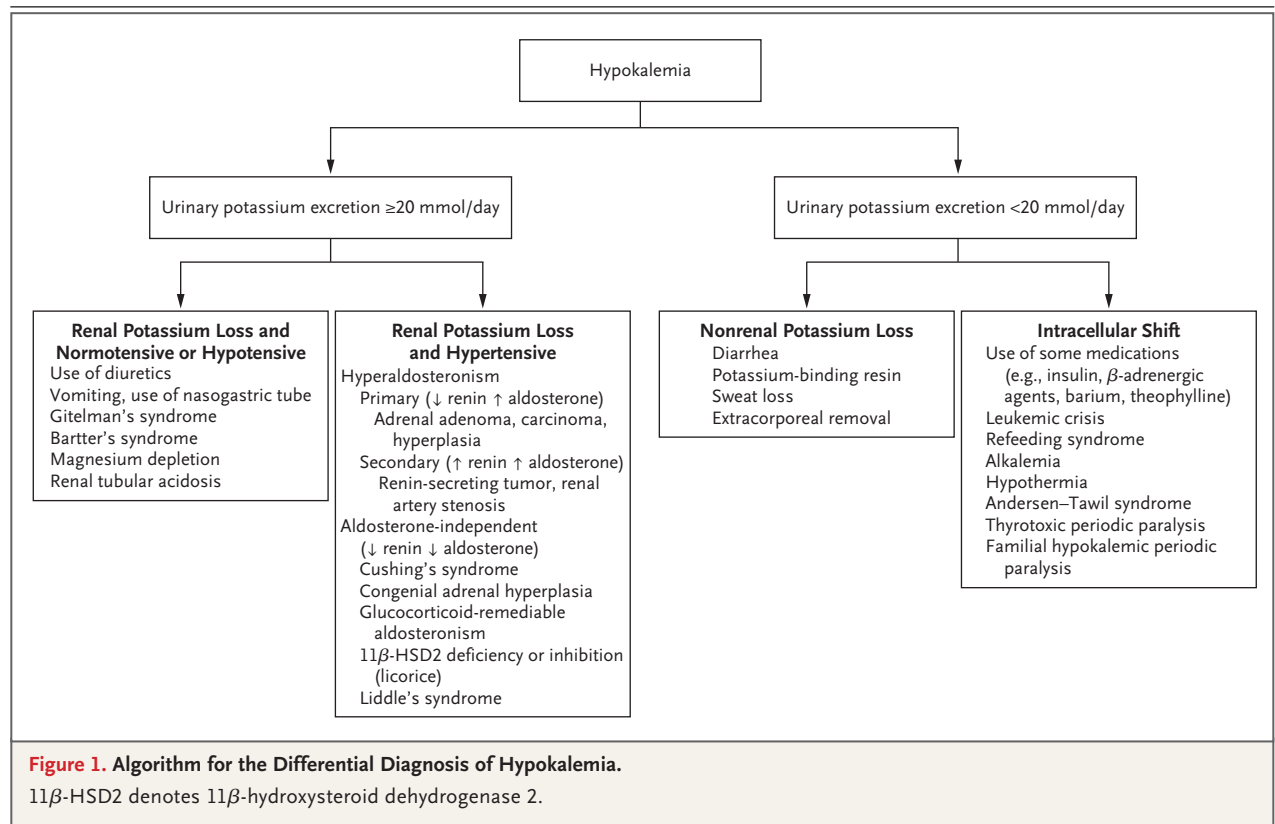
Causes of renal potassium wasting can be further characterized on the basis of the patient's blood pressure. The most common cause of renal potassium wasting associated with normotension or hypotension is the use of diuretics. Vomiting or nasogastric tube output can also cause hypokalemia due to secondary hyperaldosteronism resulting from volume contraction in addition to the loss of gastric acid in the stomach. This process leads to metabolic alkalosis and increased delivery of bicarbonate to the distal nephron, with subsequent renal excretion of bicarbonate and potassium together.

Genetic conditions such as Gitelman's syndrome or Bartter's syndrome mimic a diuretic effect, and patients generally present with hypokalemia in childhood or early adulthood. In the context of normal or low blood pressure, hypokalemia can be induced by other forms of either genetic or acquired renal tubular acidosis (i.e., acetazolamide or amphotericin exposure) and hypomagnesemia, which leads to ROMK channel activation and potassium efflux.⁷ Because

this patient was hypertensive, these diagnoses are unlikely.

Renal potassium wasting in the context of hypertension is generally mediated by the mineralocorticoid receptor. Both primary aldosteronism (characterized by a low-renin, high-aldosterone state) and secondary hyperaldosteronism (characterized by a high-renin, high-aldosterone state) can lead to hypertension and renal potassium wasting through activation of mineralocorticoid receptors by aldosterone. Primary aldosteronism can be caused by adrenal hyperplasia, adrenal adenoma, or adrenal carcinoma. Secondary hyperaldosteronism can be caused by renin-secreting tumors or renal artery stenosis. However, neither primary aldosteronism nor secondary hyperaldosteronism would explain this patient's diarrhea, tremor, and insomnia.

Mineralocorticoid receptors can also be activated by cortisol, which has a strong affinity for these receptors but is normally inactivated to cortisone by the hormone 11 β -hydroxysteroid dehydrogenase 2 (11 β -HSD2). In aldosterone-responsive epithelial cells, inhibition of 11 β -HSD2



leads to high levels of intracellular cortisol (since the enzyme resides in the cell) and activation of mineralocorticoid receptors. Other inhibitors of 11 β -HSD2 include medications (e.g., the triazole antifungal medications itraconazole and posaconazole) and certain substances in food (e.g., glycyrrhetic acid, a component of licorice).⁸ However, this patient was not known to take these medications or to eat such foods. The action of 11 β -HSD2 can be overwhelmed in patients with Cushing's syndrome or cosyntropin-secreting tumors.^{8,9} Given this patient's hypertension and proximal muscle weakness, Cushing's syndrome would be a more likely explanation if he had an inappropriately elevated urinary potassium level; however, the hallmark findings of Cushing's syndrome, such as facial plethora, supraclavicular fat pads, and abdominal striae, were not present on physical examination.

In addition, several genetic disorders are associated with apparent mineralocorticoid excess, including 11 β -HSD2 deficiency, Liddle's syndrome, congenital adrenal hyperplasia, and glucocorticoid-remediable aldosteronism. Although this patient was hypertensive, there was no known family history of hypertension, and none of these conditions would explain weight loss, diarrhea, tremor, and insomnia.

NONRENAL POTASSIUM LOSS

Excessive heat exposure and exertion can also lead to hypokalemia owing to potassium loss through the skin, although this patient reported neither of these. He reported that he had had diarrhea, which can cause gastrointestinal potassium losses as well as non-anion-gap metabolic acidosis resulting from the loss of bicarbonate in the stool. However, the occurrence of diarrhea would not explain tremor or insomnia, so I will consider other causes of hypokalemia.

INTRACELLULAR SHIFT

Persons with renal and nonrenal losses of potassium have a total body deficit of potassium. A fall in the blood level of potassium by approximately 1 mmol per liter can result in a reduction of more than 100 mmol per liter in total body potassium stores. Conversely, patients with hypokalemia that is associated with intracellular shift have normal total body potassium levels, and treatment with small amounts of potassium will normalize the blood potassium level. In this

patient, treatment with only 60 meq of potassium chloride normalized the blood potassium level, which suggests that his hypokalemia is caused by an intracellular shift of potassium. However, in such cases, the blood potassium level should be monitored in the hours and days after initial normalization in response to potassium repletion. If this patient had had low total body potassium stores, the initial rise in the blood potassium level may have been followed by a decrease as the administered potassium redistributed into the intracellular pool over time. In addition, transient hyperkalemia will develop during the repletion of potassium in one in six patients with hypokalemia.¹⁰

What could drive an intracellular shift of potassium in this patient? He was not hypothermic, and he did not use any medications that are associated with intracellular shifts of potassium, such as insulin, theophylline, or β -adrenergic agents (including decongestants containing pseudoephedrine or ephedrine). There was nothing in the patient's history that would suggest refeeding syndrome or leukemic crisis, both of which cause intracellular shifts of potassium owing to increased hematopoiesis.^{11,12} In addition, these causes of intracellular shift of potassium are all associated with mild hypokalemia. Intracellular shifts resulting in a more profound and symptomatic hypokalemia, as was seen in this patient, are more likely to be caused by familial hypokalemic periodic paralysis or thyrotoxic periodic paralysis (TPP) (Table 2).^{13,14}

Familial hypokalemic periodic paralysis is an autosomal dominant, inherited disorder caused by mutations in calcium, sodium, or potassium channels of skeletal muscles. It is most often diagnosed in patients younger than 20 years of age who are of Asian descent and have a family history of the disease. TPP is an acquired disorder resulting in increased Na⁺/K⁺-ATPase sensitivity. It occurs mostly in men of Asian descent between the ages of 20 and 50 years, although TPP has also been described in patients of Latin American descent.^{13,14} In addition, TPP can classically cause associated hypomagnesemia and hypophosphatemia, but these were not detected in this patient. A recent genomewide association study identified multiple susceptibility loci for TPP in a population of more than 2000 Chinese patients with Graves' disease (537 of whom had TPP), a finding that suggests that TPP is a molecu-

Table 2. Comparison of Thyrotoxic Periodic Paralysis and Familial Hypokalemic Periodic Paralysis.

Variable	Thyrotoxic Periodic Paralysis	Familial Hypokalemic Periodic Paralysis	This Patient
Age at onset	20 to 50 years	<20 years	27 years
Sex	More common in males than in females	Equally common in males and females	Male
Family history	Negative family history (sporadic)	Positive family history (autosomal dominant)	Aunt with a history of nonspecific weakness
Race or ethnic group	Asian (most common), Latinx (less common), White (less common)	Asian (most common)	Latinx
Genetic features	Complicated; possible molecular phenotype of Graves' disease	Mutations in calcium, sodium, or potassium channels in skeletal muscles	Unknown
Hyperthyroid features	Present	Absent	Weight loss, diarrhea, tremor, and insomnia
Electrocardiographic abnormalities	Common; possibly related to thyrotoxicosis	Uncommon	Incomplete right bundle-branch block and non-specific ST-segment abnormalities

lar subtype of Graves' disease.¹⁵ A loss of function or dominant-negative mutations in the skeletal muscle-specific potassium channel Kir2.6 may contribute to TPP, and a gain-of-function promoter variant in the encoding gene has been described in a patient with normokalemic periodic paralysis.¹⁶⁻¹⁹ Genomewide association studies have also consistently implicated a variant in *KCNJ2*, the gene^{15,20} that encodes a related skeletal muscle potassium channel, Kir2.1, which is associated with a predisposition to TPP. It is important to note that an outward-directed, inward-rectifying potassium current, mediated by Kir channels (primarily Kir2.1 and Kir2.2 tetramers), is reduced in the skeletal muscles of patients with TPP.²¹

Clinical episodes in patients with TPP and familial hypokalemic periodic paralysis consist of intermittent and often progressive proximal muscle weakness and even paralysis in the context of profound hypokalemia. Episodes commonly occur after high carbohydrate intake (which results in endogenous insulin release, further driving potassium into cells by means of the Na⁺/K⁺-ATPase); strenuous exercise (from a rebound influx of potassium back into cells following skeletal muscle release during physical activity); or as in this case, on awakening in the morning.

Although this patient had a maternal aunt who had had a nonspecific episode of weakness, I would expect a history of more repeated episodes

if there were a true family history of periodic paralysis. In addition, this patient's age is more consistent with TPP than with familial hypokalemic periodic paralysis. It is important to note that his weight loss, tachycardia, hypertension, conjunctival injection, and tremor suggest hyperthyroidism, which favors a diagnosis of TPP. Moreover, electrocardiographic abnormalities are more commonly found in patients with TPP than in those with familial hypokalemic periodic paralysis, possibly owing to the cardiac effects of thyrotoxicosis. In this case, an electrocardiogram showed no U waves (small deflections that follow the T waves), but there was evidence of right bundle-branch block. Although clinical episodes are similar in patients with familial hypokalemic periodic paralysis and those with TPP, distinguishing between the two diagnoses is important for treatment and prognosis. TPP can be functionally cured with the correction of hyperthyroidism, whereas familial hypokalemic periodic paralysis often results in progressive muscle weakness despite the avoidance of triggers.

OTHER DIAGNOSTIC POSSIBILITIES

The patient had mixed respiratory and non-anion-gap metabolic acidosis, which is not explained by a diagnosis of TPP. His non-anion-gap metabolic acidosis is likely related to the diarrhea, which leads to an increased loss of bicarbonate in the gastrointestinal tract. The respiratory acidosis

could have been caused by respiratory muscle weakness in the context of hypokalemia. Alternatively, the sedating effects of substances could have led to respiratory acidosis. This patient was taking an herbal supplement (Nervitran) that contains valerian, passionflower, and hops and is marketed in Latin America as a treatment for stress, anxiety, and insomnia. I could not find any published case reports associating this product with hypokalemia, but each of these components are postulated to have sedative effects, possibly through interaction with γ -aminobutyric acid receptors.

Several other laboratory abnormalities are left unexplained by TPP, including a mildly elevated aldolase level and elevated levels of aspartate aminotransferase and alanine aminotransferase. Perhaps these findings are due to a low degree of muscle-cell death or the hepatotoxic effect of either alcohol or supplement intake.

Overall, this 27-year-old man's presentation with subacute symptoms that were suggestive of hyperthyroidism and symptomatic hypokalemia, which occurred after physical exertion and responded to modest potassium repletion, is most consistent with a diagnosis of TPP. I would confirm the diagnosis with tests of thyroid function. Given the association between TPP and Graves' disease, I would also test for the presence of activating thyrotropin-receptor autoantibodies.

DR. ANDREW S. ALLEGRETTI'S DIAGNOSIS

Thyrotoxic periodic paralysis.

INITIAL DIAGNOSTIC TESTING

Dr. Peter M. Sadow: The level of thyrotropin was fully suppressed (<0.1 μ IU per milliliter; reference range, 0.4 to 5.00). On the basis of the thyrotropin result, a blood specimen was obtained to measure thyroid function. Tests showed markedly increased levels of free thyroxine (free T4), at 6.7 ng per deciliter (reference range, 0.9 to 1.8), and total triiodothyronine (T3), at 525 ng per deciliter (reference range, 60 to 181). A blood specimen was sent to a central reference laboratory for evaluation. The thyroid-stimulating immunoglobulin test index was 4.6 (reference

value, ≤ 1.5) and the thyrotropin-binding inhibitory immunoglobulin level 26 IU per liter (reference range, 0.00 to 1.75), a value that is 99% specific for Graves' disease.

DISCUSSION OF MANAGEMENT

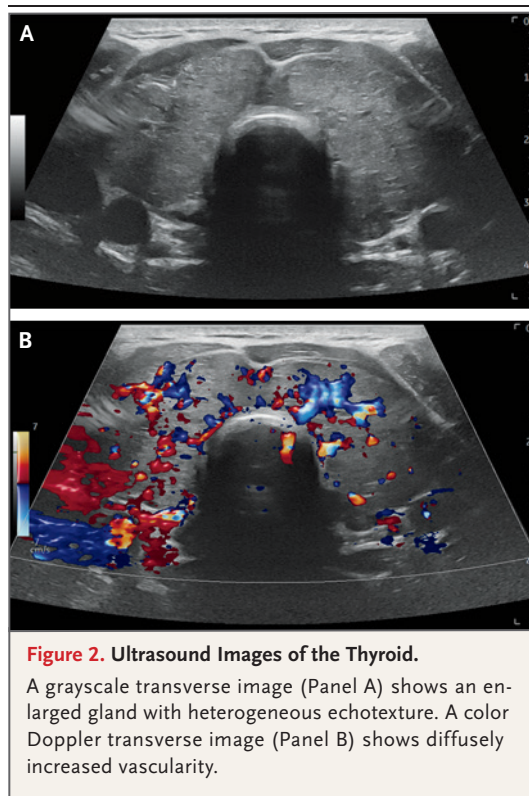
Dr. Nikolaos Stathatos: Because hypokalemia in patients with TPP is related to intracellular shift of potassium, there is a risk of overcorrection that can result in fatal hyperkalemia.²²⁻²⁴ Therefore, it is recommended that potassium replacement be limited to no more than about 90 meq in the first 24 hours or less than 10 meq per hour.²² This patient received a total of 110 meq of potassium chloride in the first 24 hours, with a peak blood potassium level of 5.2 mmol per liter that subsequently normalized.

Thyroid hormone directly increases the activity of Na^+/K^+ -ATPase, but it also stimulates it at the gene level by increasing its transcription. Adrenergic activity, which is increased in the context of hyperthyroidism, also stimulates Na^+/K^+ -ATPase.²⁵ Initially, we counseled the patient about the elimination of precipitating factors, including carbohydrate-rich foods (which can lead to hyperinsulinemia), catecholamine surges, and alcohol intake. Ultimately, hypokalemia in patients with TPP is quickly reversible with the correction of hyperthyroidism.

The patient was treated with methimazole and propranolol. Some experts recommend upfront use of propranolol to correct the hypokalemia associated with TPP.²⁶ He had resolution of hypokalemia and clinical symptoms within 24 hours after the initiation of treatment. Imaging studies were obtained.

Dr. Cynthia L. Czawlytko: Grayscale ultrasound imaging of the thyroid revealed a diffusely enlarged gland with heterogeneous echotexture (Fig. 2). Color Doppler imaging showed diffusely increased vascular flow. Imaging findings were compatible with a clinical diagnosis of Graves' disease.

Dr. Stathatos: The patient was followed closely after he was discharged from this hospital. The blood levels of free T4 and total T3 decreased and were close to or within the normal range within 3 weeks after discharge. During that time, his potassium level remained within the normal



range, with values between 3.5 and 5.0 mmol per liter.

Given the risk of recurrence of life-threatening hypokalemia, the patient elected to proceed with total thyroidectomy as a permanent solution. He underwent surgery 39 days after his initial presentation.

PATHOLOGICAL DISCUSSION

Dr. Sadow: The excised thyroid gland weighed 52 g (upper limit of the normal range, 25), and the right lobe was slightly larger than the left lobe. The gross appearance was uniform bilaterally, with a vague whorled appearance that varied in color from red to tan to pink. Microscopic evaluation revealed scattered lymphoid aggregates with a histologically nodular growth pattern and papillary hyperplastic projections into colloid (Fig. 3). Areas with cleared-out colloid and a mixed pattern of peripheral macrofollicles and compressed microfollicles are diagnostic of partially treated Graves' disease, given the mixture of empty follicles and normal macrofollicles.

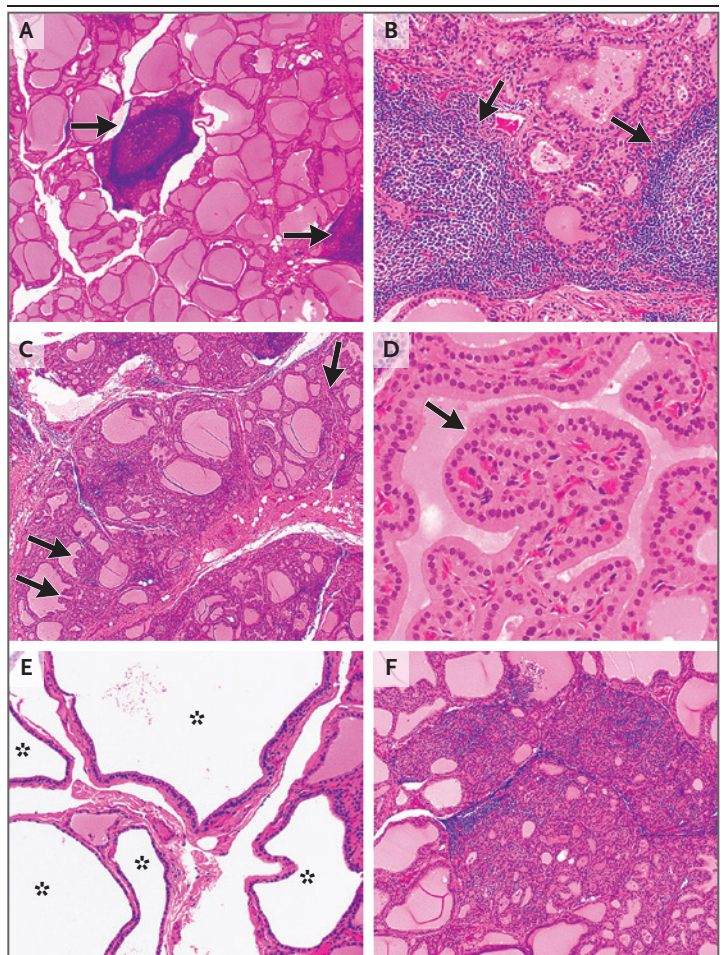


Figure 3. Resection Specimen of the Thyroid.

Hematoxylin and eosin staining of sections of a specimen of the excised thyroid gland show lymphoid aggregates with germinal centers (Panels A and B, arrows) that are the hallmark of chronic thyroiditis. At a lower magnification (Panel A), all the follicles are macrofollicles that are typically seen (except for the lymphoid aggregates) in normal thyroid tissue. Graves' disease is evidenced by the hyperplasia, which appears in a nodular pattern with papillary architectural projections into the colloid-rich follicles (Panel C, arrows). These papillary projections (Panel D, arrow) have fibrovascular cores and round, normochromatic nuclei. Empty follicles (Panel E, asterisks) indicate excreted colloid after stimulation by thyroid-stimulating immunoglobulin. The mixed pattern of compressed, empty follicles (Panel F, center) and normal macrofollicles (Panel F, periphery) is indicative of effective medical treatment.

PATHOLOGICAL DIAGNOSIS

Diffuse nodular follicular hyperplasia with papillary architecture consistent with partially treated Graves' disease.

ADDITIONAL MANAGEMENT

Dr. Stathatos: After surgery, treatment with levothyroxine replacement was initiated. Since then, the patient has remained asymptomatic and has resumed all his daily activities. At the time of his last follow-up, the blood levels of thyrotropin and free T4 were normal.

FINAL DIAGNOSIS

Thyrotoxic periodic paralysis associated with Graves' disease.

This case was presented at the Medicine Case Conference, Interview Series.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank Dr. Eugene Rhee for his review of the case presentation.

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